PRINCETON UNIVERSITY
Electrical & Computer Engineering

What Matters for Policy Distillation in POMDPs

Teacher Quality, Memory Mechanism, and Architecture Choice

Mayan Wasu · Class of 2027 | **Advisor:** Prof. Hossein Valavi | mayanwasu@princeton.edu

5× ARCHITECTURES | **8×** TEACHER LEVELS | **16.3×** EDGE SPEEDUP | READ PAPER & CODE

TL;DR Across **500+ distillation experiments**, teacher quality explains 49–72% of student return variance — **4–16× more** than student architecture. But architecture matters *more* when teachers are bad (17–25% interaction). Pick a good teacher first; then match architecture to task: **GRU** for locomotion, **Transformer** for content-addressable recall, and **STU / Transformer** for spatially-structured driving (where GRU drops to mid-pack).

ARCHITECTURES | GRU *locomotion* | LSTM *retention* | Transformer *recall* | STU *spatial · LDS* | STU-T *tensordot variant* | MLP *no memory*

SAME COLORS USED IN EVERY CHART →

MOTIVATION & SETUP 01

§1.1 THE PROBLEM

Deploying RL agents on robots and drones requires compressing large **"teacher" policies** into tiny **"student" models** via behavioral cloning. The student only sees **partial observations** — joint positions, no velocities — and must *infer missing state from history*. Memory is no longer optional.

RESEARCH QUESTION
Should you invest your engineering effort in a smarter student architecture — or in a better teacher?

§1.2 DISTILLATION PIPELINE

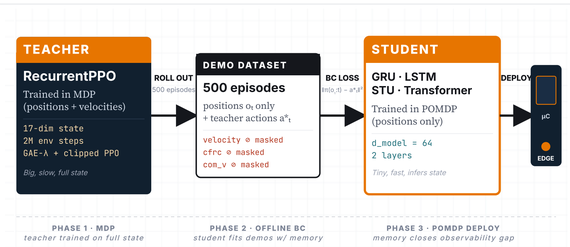

Figure 1. Three-phase pipeline. A fully-observable RecurrentPPO teacher rolls out 500 demonstration episodes; velocities are stripped before behavioral cloning so the student must close the observability gap with memory.

§1.3 ARCHITECTURES

ARCHITECTURE	PARAMS	LATENCY	MEMORY MECHANISM
GRU	51 K	1.26 ms	Learned gating
LSTM	68 K	1.33 ms	Gating + cell state
Transformer	232 K	3.36 ms	Attention lookup
STU (Spectral)	338 K	4.38 ms	Fixed Hankel filters
MLP (no memory)	5 K	0.10 ms	None

All recurrent students sized to `d_model = 64`*, 2 layers — a compact, edge-feasible regime.*

How each architecture remembers · cartoon view

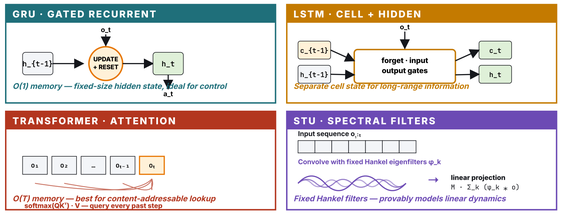

Figure 2. Each architecture maintains memory differently. **GRU/LSTM** compress history into a fixed hidden state; **Transformer** attends to the entire past at every step; **STU** convolves with fixed spectral filters and projects.

§1.3b BRIEFLY · WHAT IS PPO?

PPO (Proximal Policy Optimization) trains the *teacher*: roll out π_θ, then update toward high-advantage actions while *clipping* so the new policy can't drift too far. **RecurrentPPO** adds a GRU/LSTM so the teacher itself can solve POMDPs — yielding strong, history-aware demos for the student to imitate.

§1.4 ENVIRONMENTS & PROTOCOL

A MuJoCo POMDP
HalfCheetah & Ant — velocity channels removed; 8 teacher quality levels.

B MetaDrive
Autonomous driving with 72-laser lidar (91-dim); rich spatial observations.

C T-Maze
Synthetic *retention* — hold a 1-bit hint over $L \in \{10,50,100,200\}$ steps.

D CopyTask
Synthetic *recall* — replay $N \in \{5,10,20\}$ tokens after a delay.

DEMOS	SEEDS	TEACHER LEVELS	D_MODEL
500 episodes / cond.	3 – 5 student seeds	8 200K → 2M ± noise	64 edge-feasible

§1.6 CONTRIBUTIONS

1. **Variance decomposition** across 500+ runs: teacher $\eta^2 \approx$ **50–72%** vs architecture $\eta^2 \approx$ **5–14%**.
2. **Memory taxonomy** separating *retention, recall,* and *state inference* — each with a different architectural winner.
3. **Flash-STU tensordot regression** at `d=64`: 52× worse MSE than full STU; effectively random on every distillation task. *Unreported in prior work.*
4. **JEPA null result** across 4 archs × 3 envs and a 6-value λ sweep on LDS.
5. **SpectraLDS deployment path: 16.3× edge speedup** over full STU at 0.04% reconstruction error.

§1.7 HOW IT WORKS · THREE-PHASE PIPELINE

Phase 1 — Train a powerful teacher. We train a large RecurrentPPO agent with *full* access to the environment (positions + velocities). It learns through millions of timesteps of trial and error. This teacher is too big for real-time deployment, but it knows how to act.

Phase 2 — Record and compress. We record 500 episodes of the teacher acting. Crucially, we *strip the velocity channels* — the student only sees joint positions. The student trains via behavioral cloning: minimize the squared error between its actions and the teacher's at every timestep.

$$L_{BC}(\theta) = \Sigma_{episodes}\ \Sigma_{t=1..T} \parallel \pi_\theta(o_{1:t}) - a^*_t \parallel^2$$

Longer episodes contribute more gradient signal. Successful (longer) episodes are implicitly upweighted — the student learns more from the teacher's best behavior.

Phase 3 — Deploy at the edge. The tiny student (51K-338K params) runs in real-time. It sees only positions and must **infer velocities from past observations** — this is why temporal memory matters.

Key insight: Distillation is *supervised learning,* not RL. The teacher already explored; the student just fits a function. This is why teacher quality matters more than architecture.

Acknowledgements. I thank **Prof. Hossein Valavi** for his guidance throughout this project, and the **Princeton Research Computing** group for access to the Della and Adroit clusters.

CENTRAL FINDING · TEACHER DOMINATES 02

THE HEADLINE FINDING

49–72% of variance in student return is explained by **teacher quality — 4–16× more** than architecture choice ($\eta^2 \approx$ 5–14%). *Pick the teacher first.*

TEACHER H^2 (VARIANCE EXPLAINED)
61%
pooled across MuJoCo + MetaDrive

ARCHITECTURE H^2
9%
GRU vs LSTM vs Transformer vs STU

§2.1 STUDENTS TRACK TEACHERS

Student return vs. teacher return — MuJoCo POMDPs

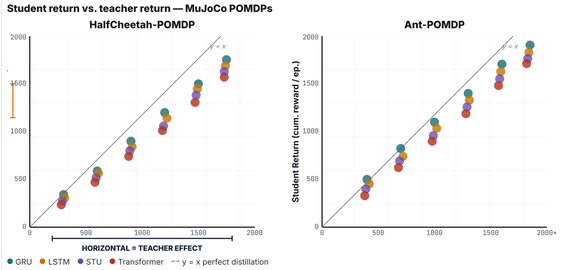

Figure 3. Each marker is one (architecture, teacher) condition averaged over 3 seeds. **Horizontal spread (teacher) > vertical spread (architecture).** Architecture choice rotates the cluster but never crosses lanes.

> "The teacher is the signal; the architecture is just the antenna."

§2.2 VARIANCE DECOMPOSITION (ANOVA, H^2)

ENVIRONMENT	TEACHER	ARCHITECTURE	INTERACTION
HalfCheetah	71.9%	4.6%	17.3%
Ant	49.1%	11.0%	25.2%
MetaDrive	58.3%	14.1%	19.7%

Teacher η^2 is 4–16× larger than architecture η^2. The interaction term (17–25%) tells us *when* architecture matters — it grows precisely when teachers are weak.

§2.3 TEACHER QUALITY SWEEP · REAL NUMBERS

Ant-POMDP · 8 teacher conditions × 4 students (3 seeds)

TEACHER (STEPS · NOISE)	TEACH.	GRU	LSTM	STU	TRANSF.
200K · clean	647	894	847	861	879
200K · σ=0.3	216	961	952	883	961
500K · clean	963	1112	1121	1050	1068
500K · σ=0.3	251	1079	1161	1083	1098
1M · clean	1424	1745	1692	1356	1657
1M · σ=0.3	317	1145	945	1134	962
2M · clean	439	1783	1238	496	858
2M · σ=0.3	210	1921	1944	1359	1091

Table 1. GRU consistently leads or co-leads. Students *routinely exceed* imperfect teachers because BC averages over a high-variance demo set with rare strong rollouts.

§2.4 WHY STUDENTS EXCEED TEACHERS

The Ant 2M teacher scores **439**; demos range **17 → 1928** (σ=368, ~15% of demos > 800). BC's per-step MSE *implicitly upweights longer, more successful episodes* — the student inherits a return-biased distribution and generalizes beyond it.

1.23× **median student / teacher ratio** with stable teachers; ratio grows with teacher variance.

§2.5 CROSS-ENVIRONMENT MEAN RETURN

Overall mean return across MuJoCo POMDPs (all teachers, 3 seeds)

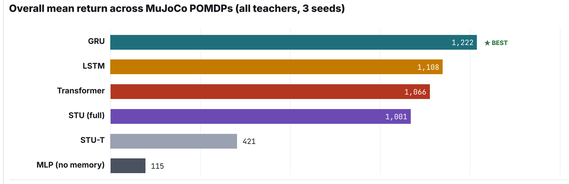

Figure 4. Pooled across all teacher levels and MuJoCo envs. The recurrent-vs-spectral gap is small (1,222 vs 1,001); the **memory-vs-no-memory** gap is the order of magnitude (1,222 vs 115).

§2.6 PARAMETER-MATCHED COMPARISON

Is GRU's lead just a parameter advantage? No. We re-ran GRU at 312K params (matched to STU's 338K) on Ant.

MODEL	PARAMS	ANT MEAN	BEST RUN
GRU (compact)	51 K	1,222	1,921
GRU (matched)	312 K	1,422	2,354
STU (full)	338 K	1,001	1,657

GRU's advantage is genuinely architectural — and scales with capacity. The matched-param GRU set the **highest return in the entire study** (2,354 on Ant).

WHEN ARCHITECTURE MATTERS 03

§3.1 MEMORY MECHANISM TAXONOMY

THREE KINDS OF MEMORY · THREE WINNERS

Figure 5. Each architecture wins a different memory operation. There is *no universal best.*

§3.2 SYNTHETIC PROBES · HARD NUMBERS

T-Maze · success rate (1.0 = perfect, 3 seeds)

L (GAP)	GRU	LSTM	TRANSF.	STU-T
10	1.00	1.00	1.00	0.05
50	1.00	1.00	1.00	0.05
100	1.00	1.00	1.00	0.05
200	1.00	1.00	0.05	0.05

Figure 6. GRU/LSTM hold the hint perfectly across all gaps; **Transformer collapses at L=200** as the corridor exceeds its window. STU-T fails everywhere — random.

CopyTask · score vs N (max = N, optimal teacher)

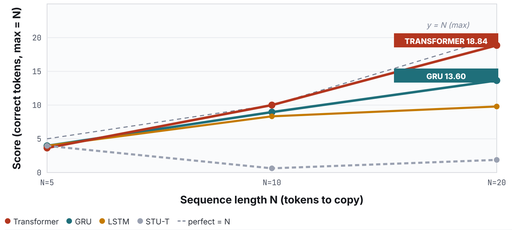

Figure 7. As *N* grows, attention pulls away. **Transformer 18.84/20** at N=20 — content-addressable lookup is the right primitive for recall.

§3.3 DATA EFFICIENCY · COPYTASK N=10

Score vs demo budget (3 seeds)

BUDGET	GRU	LSTM	TRANSF.	STU-T
5	1.41	1.41	1.31	1.10
50	2.21	2.32	2.54	1.07
250	5.30	4.83	6.87	1.19
1000	9.00	7.85	9.00	1.16

Figure 8. Even **5 demos** get GRU/LSTM to ≈30% of perfect on a recall task. Transformer overtakes once budget ≥ 50.

§3.4 LDS REPLICATION · SPECTRAL MODELS

MSE vs spectral radius (16 seeds, lower is better)

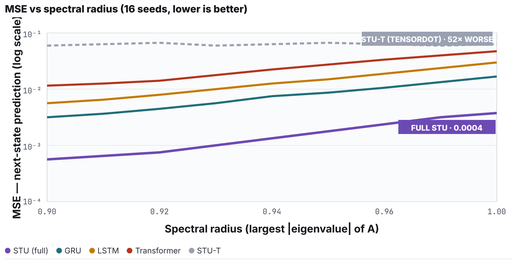

Figure 9. UNREPORTED Flash-STU's tensordot variant is **52× worse than published** at `d=64`. Full STU is **12× better than GRU**.

Practical contraindication: below `d_model = 128`, the "efficient" tensordot variant is essentially random on every distillation task — surfaced here for the first time.

§3.5 ARCHITECTURE × TASK WINNER MATRIX

Per-task best architecture (green = top-1, amber = within 5%)

	GRU 51 K	LSTM 68 K	Transf. 232 K	STU 338 K	MLP 5 K
HalfCheetah	★ 1,422	1,238	858	496	−180
Ant	★ 1,222	1,108	1,066	1,001	115
MetaDrive	61.5	70.0	73.7	★ 75.4	16.4
T-Maze (L=200)	★ 1.00	★ 1.00	0.05	0.05	0.05
CopyTask (N=20)	13.60	10.48	★ 18.84	1.16	1.04
LDS replication	5.0e-3	2.6e-3	1.0e-3	★ 4e-4	2.1e-2

Figure 10. No architecture wins everywhere. **GRU dominates locomotion + retention**; **STU dominates spatial driving + LDS**; **Transformer dominates recall**. The right tool depends on the memory operation, not the field.

DRIVING · DEPLOYMENT · RECOMMENDATIONS 04

§4.1 METADRIVE · SPATIAL STORY FLIPS

On **MetaDrive** — autonomous driving with 72-laser lidar (91-dim) — the locomotion ranking *flips*. Recurrent students struggle through complex intersections; **STU and Transformer** exploit spatial geometry directly.

Mean episode reward (points / episode) · MetaDrive (5 seeds)

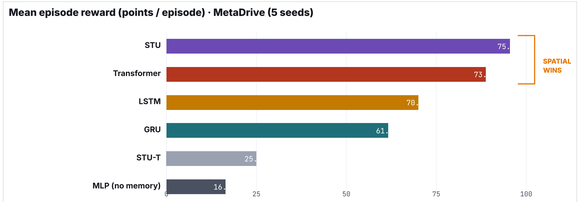

Figure 11. STU (75.4) and Transformer (73.7) lead on lidar-rich MetaDrive; GRU (68.7) drops to mid-pack — observation structure changes the right tool.

Rollout traces · 6 architectures × 5 scenarios

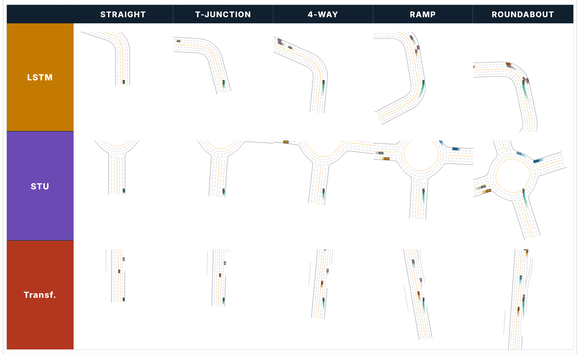

Figure 12. Three key architectures across five road types. **STU and Transformer** negotiate branches and roundabouts; **LSTM** hesitates at complex geometry.

§4.2 BEST ROLLOUTS · MUJOCO

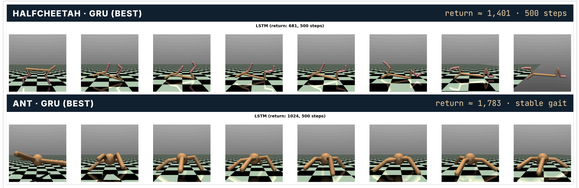

Figure 13. Best-of-sweep students. GRU develops stable gaits on both bodies; worst students (Transformer on HalfCheetah, STU on Ant) stall or fall — see paper appendix.

§4.3 JEPA AUXILIARY LOSS · NULL

We tested a **JEPA** predictive auxiliary loss (predict teacher's future hidden state) across all architectures, environments, and a 6-value λ sweep. **No consistent benefit found.** MSE = 0.1053 at every λ; MuJoCo Δ ranges −4.5% to +5.7%. *Skip JEPA for policy distillation.*

§4.4 EDGE DEPLOYMENT

Inference latency — milliseconds per environment step (lower = better)

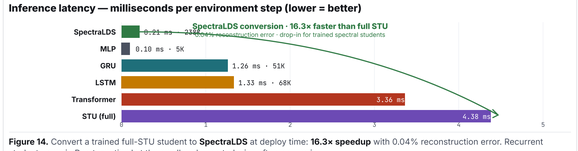

Figure 14. Convert a trained full-STU student to **SpectraLDS** at deploy time: **16.3× speedup** with 0.04% reconstruction error. Recurrent students remain Pareto-optimal at the small end; spectral wins after conversion.

PRACTICAL RECOMMENDATIONS

What to do

- 01 **Invest in teacher quality first.** 4–16× more impact than the architecture choice.
- 02 **GRU for locomotion & control.** 51K · 1.26 ms — Pareto-optimal in our sweep.
- 03 **Transformer for content recall.** If the task is "remember which," use attention.
- 04 **STU / Transformer for spatial inputs** — lidar, radar, occupancy grids.
- 05 **Use full STU below d_model = 128.** The "efficient" tensordot path is silently random.
- 06 **Skip JEPA auxiliary losses.** No consistent benefit found.
- 07 **Convert to SpectraLDS at deploy.** 16.3× speedup for spectral students.

§4.6 LIMITATIONS · FUTURE WORK

LIMITATIONS
- Single MuJoCo teacher seed (42).
- 3 student seeds (5 on key conditions).
- Compact only — `d_model=64`.
- Param-matched: GRU_{312K} (312K) outperforms STU (338K).
- Offline BC; DAgger untested.
- SpectraLDS not tested end-to-end on a distilled student.

FUTURE
- Episode-length-normalized BC ablation.
- Multi-seed teachers.
- `d_model` sweep 64 → 256.
- Pixel-based POMDPs.
- Mamba state-space baseline.